Statistical Testing & Classification Reference

Updated Pipeline — Detection and Attribution of AI-Generated and Human-Written Phishing Emails

Mannan Bhardwaj (25BYB0013) Bhoumik Singh Rajput (25BYB0120)

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Scope of This Document

This document explains every statistical test and classification model in the *current* version of the pipeline — the version with the normality-testing branch removed (Kruskal–Wallis is used for all continuous features, unconditionally) and the two-group Chi-Square test added for categorical attribution features. It does not describe earlier design iterations. For each test: what it does, when it runs, its formula, a fully worked numeric example, and how to read the result. The last section walks one example email through the entire pipeline end to end.

1 Quick Reference

The first seven rows run **once**, during training, to decide which features are trustworthy. The last three rows run **every time** a new email needs to be classified, and never involve a hypothesis test or a p-value.

2 Phase 1: Feature-Selection Tests

These four tests are run once on the training set. Each produces a p-value per feature; none of them ever run again once the feature list is locked.

2.1 Kruskal–Wallis H Test

Purpose: Tests whether a continuous feature's values differ across the three classes (Legitimate, Human-Phishing, AI-Phishing), without assuming the data is normally distributed.
When it runs: On every continuous feature, comparing all three classes at once.

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1)$$

where N is the total sample size, k is the number of classes, R_j is the sum of ranks in class j, and n_j is the size of class j.

Name	Purpose	Feature type	Runs on
Kruskal–Wallis H	Does a feature differ across the 3 classes?	Continuous	Legit / Human / AI
Chi-Square (3-group)	Does a flag depend on class?	Categorical	Legit / Human / AI
Mann–Whitney U	Does a feature separate the two phishing types?	Continuous	Human vs. AI only
Chi-Square (2-group)	Does a flag separate the two phishing types?	Categorical	Human vs. AI only
Benjamini–Hochberg FDR	Corrects all p-values for multiple testing	All	—
Cohen's d	How large is a continuous difference, really?	Continuous	—
Cramér's V	How strong is a categorical association, really?	Categorical	—
Model 1: Rank-Binned NB	Classifies using verified continuous features	Continuous	Every new email
Model 2: Categorical NB	Classifies using verified categorical features	Categorical	Every new email
Meta-Naive Bayes	Combines Model 1 & Model 2 into one verdict	—	Every new email

Worked example — feature: sentence perplexity variance ("burstiness"). Four emails sampled from each class:

Legitimate	2	3	4	5
Human-Phishing	9	10	11	12
AI-Phishing	5	6	7	8

Pool all 12 values and rank them 1 (lowest) to 12 (highest). The value 5 appears twice (once in Legitimate, once in AI-Phishing), so both instances share the average of ranks 4 and 5, i.e., rank 4.5.

Class	Ranks	Rank sum R_j
Legitimate	1, 2, 3, 4.5	$R_1 = 10.5$
Human-Phishing	9, 10, 11, 12	$R_2 = 42$
AI-Phishing	4.5, 6, 7, 8	$R_3 = 25.5$

With $N = 12$, $k = 3$, $n_j = 4$ for every class:

$$H = \frac{12}{12 \times 13}\left(\frac{10.5^2}{4} + \frac{42^2}{4} + \frac{25.5^2}{4}\right) - 3(13) = 0.0769 \times 631.125 - 39 \approx 9.55$$

Decision rule: Compare H to a chi-square distribution with $k - 1 = 2$ degrees of freedom. The critical value at $p = 0.05$ is 5.99. Since $9.55 > 5.99$, the difference is statistically significant — burstiness genuinely differs across the three classes, and the feature proceeds to the FDR/effect-size check.

2.2 Chi-Square Test (Three-Group)

Purpose: Tests whether a categorical (yes/no) feature is associated with class membership.
When it runs: On every categorical/flag feature, comparing all three classes at once.

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed count in cell (i, j) and $E_{ij} = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$ is the count expected if the feature were unrelated to class.

Worked example — feature: contains an urgency keyword (Yes/No), 100 emails per class.

	Yes	No	Total
Legitimate	10	90	100
Human-Phishing	70	30	100
AI-Phishing	55	45	100
Total	135	165	300

Expected count per cell (e.g., Legitimate/Yes) $= \frac{135 \times 100}{300} = 45$; every class's expected Yes/No split is 45/55 since class is assumed irrelevant. Summing $(O - E)^2/E$ over all six cells gives $\chi^2 \approx 78.8$.
Decision rule: With df $= (3 - 1)(2 - 1) = 2$, the critical value at $p = 0.05$ is 5.99. $78.8 \gg 5.99$: urgency-keyword usage is strongly tied to class.

2.3 Mann–Whitney U Test (Attribution)

Purpose: Tests whether a continuous feature separates AI-Phishing from Human-Phishing specifically — a different, narrower question than the three-group tests above.
When it runs: On every continuous feature, restricted to the phishing-only subset (Human-Phishing vs. AI-Phishing).

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Worked example — feature: urgency-keyword count per email.

Human-Phishing	6	7	8	9	10
AI-Phishing	2	3	3	4	5

Pooling and ranking all 10 values (the tied pair of 3's each take the average rank 2.5): Human-Phishing ranks are $6, 7, 8, 9, 10 \to R_1 = 40$; AI-Phishing ranks are $1, 2.5, 2.5, 4, 5 \to R_2 = 15$.

$$U = (5)(5) + \frac{5(6)}{2} - 40 = 25 + 15 - 40 = 0$$

Decision rule: U ranges from 0 (complete separation) to $n_1 n_2 = 25$ (complete overlap). $U = 0$ here means every Human-Phishing email outranked every AI-Phishing email on this feature — the strongest possible signal, and a very small p-value.

2.4 Chi-Square Test (Two-Group, Attribution)

Purpose: The categorical counterpart to Mann–Whitney U: tests whether a categorical/flag feature separates AI-Phishing from Human-Phishing specifically.
When it runs: On every categorical feature, restricted to the phishing-only subset.
Same formula as Section 2.2, but with only two classes (rows) instead of three.

Worked example — feature: contains a suspicious link (Yes/No), 100 emails per phishing type.

	Yes	No	Total
Human-Phishing	60	40	100
AI-Phishing	25	75	100
Total	85	115	200

Expected Yes per group $= \frac{85 \times 100}{200} = 42.5$; expected No $= 57.5$. Summing $(O - E)^2/E$ over all four cells gives $\chi^2 \approx 25.08$.
Decision rule: With df $= 1$, the critical value at $p = 0.05$ is 3.84. $25.08 \gg 3.84$: link presence is strongly tied to *which type* of phishing an email is, not just whether it's phishing at all.

3 Multiple-Testing Correction and Effect Size

Every test above only tells you whether a difference is *probably real*. These two checks decide whether it is *real and worth keeping*.

3.1 Benjamini–Hochberg FDR Correction

Purpose: When many features are tested at once, some will look "significant" by pure chance. This correction controls the expected proportion of false positives among everything ultimately kept.
When it runs: Once, across every raw p-value produced by every test above, jointly.

$$P_{(i)} \leq \frac{i}{m} Q \qquad (Q = 0.05)$$

Sort all m p-values ascending and rank them $i = 1, \ldots, m$. Find the largest i whose p-value is still below its own threshold $\frac{i}{m}Q$; keep every feature ranked at or below that i.

Worked example — 10 features tested.

Rank i	p-value	Threshold $\frac{i}{10}(0.05)$	Passes?
1	0.001	0.005	✓
2	0.004	0.010	✓
3	0.006	0.015	✓
4	0.011	0.020	✓
5	0.019	0.025	✓
6	0.022	0.030	✓
7	0.030	0.035	✓
8	0.040	0.040	✓
9	0.060	0.045	×
10	0.080	0.050	×

Decision rule: The largest passing rank is 8, so features ranked 1–8 are kept; ranks 9–10 are discarded, even though a naive uncorrected 0.05 cutoff would have kept more of them.

3.2 Cohen's d

Purpose: Measures how large a continuous feature's difference between two groups actually is, in standardized units — protects against large sample sizes making trivial differences look "significant."
When it runs: On every continuous feature that passed its p-value test, for every pairwise comparison needed.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}, \qquad s_{\text{pooled}} = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$$

Rule of thumb: $|d| \approx 0.2$ small, 0.5 medium, 0.8 large. This pipeline requires $|d| \geq 0.20$.

Worked example — feature: average sentence length. Legitimate: $12, 14, 13, 15, 16$ (mean 14); Phishing: $15, 17, 16, 14, 18$ (mean 16). Both groups have standard deviation ≈ 1.58, so pooled std ≈ 1.58.

$$d = \frac{16-14}{1.58} \approx 1.27$$

Decision rule: 1.27 is a large effect — Phishing emails use noticeably longer sentences than Legitimate ones, not just detectably-but-trivially longer ones.

3.3 Cramér's V

Purpose: The categorical counterpart to Cohen's d: measures how strong a categorical association actually is, on a 0-to-1 scale.
When it runs: On every categorical feature that passed its p-value test.

$$V = \sqrt{\frac{\chi^2}{n \times (\min(\text{rows},\text{cols}) - 1)}}$$

Rule of thumb (1 degree of freedom): $V \approx 0.1$ weak, 0.3 moderate, 0.5+ strong.

Worked example — reusing the suspicious-link feature from Section 2.4 ($\chi^2 = 25.08$, $n = 200$, df $= 1$).

$$V = \sqrt{\frac{25.08}{200 \times 1}} = \sqrt{0.1254} \approx 0.354$$

Decision rule: A moderate-to-strong association — link presence is a genuinely useful attribution signal, not just a statistical technicality.

4 Phase 2: Classification Models

These are not hypothesis tests and do not produce p-values. They run for every new email, using only the parameters (bin edges, frequency tables, class priors) that Phase 1 already computed and locked. Nothing here is recalculated per email.

4.1 Model 1: Rank-Binned Naive Bayes

Purpose: Classifies using continuous features that survived Kruskal–Wallis, by binning each feature's value into intervals (e.g., Low/Medium/High) and looking up how often each class historically fell into that bin.

$$P(x_i \in \text{Bin}_b \mid C_k) = \frac{N_{b,k} + \alpha}{N_k + \alpha B}, \qquad \alpha = 1 \text{ (Laplace smoothing)}$$

Worked example — feature: burstiness, using training counts (1,000 emails/class).

	Low	Medium	High
Legitimate	700	250	50
Human-Phishing	200	400	400
AI-Phishing	650	300	50

A new test email's burstiness falls in the **High** bin:

$$P(\text{High} \mid \text{Legit}) = \frac{51}{1003} \approx 0.051, \quad P(\text{High} \mid \text{Human}) = \frac{401}{1003} \approx 0.400, \quad P(\text{High} \mid \text{AI}) = \frac{51}{1003} \approx 0.051$$

4.2 Model 2: Categorical Naive Bayes

Purpose: Classifies using categorical features that survived Chi-Square, by looking up how often each class historically showed that category value.

$$P(x_i = v \mid C_k) = \frac{\text{Count}(x_i = v \text{ in } C_k) + \alpha}{\text{Total in } C_k + \alpha K}$$

Worked example — feature: urgency keyword present (Yes/No), 1,000 emails/class.

	Yes	No
Legitimate	50	950
Human-Phishing	700	300
AI-Phishing	550	450

The same test email contains an urgency keyword (**Yes**):

$$P(\text{Yes} \mid \text{Legit}) \approx 0.051, \quad P(\text{Yes} \mid \text{Human}) \approx 0.700, \quad P(\text{Yes} \mid \text{AI}) \approx 0.550$$

4.3 Meta-Naive Bayes Probability Combiner

Purpose: Combines Model 1's and Model 2's independent judgments into one final probability per class, using Bayes' theorem — not a simple average, so confident agreement between the two models is amplified and disagreement is suppressed.

$$P(C_k \mid P_1, P_2) = \frac{P(C_k)\, P_1(C_k)\, P_2(C_k)}{\sum_j P(C_j)\, P_1(C_j)\, P_2(C_j)}$$

With equal priors $P(C_k) = \frac{1}{3}$:

Legit: $\frac{1}{3}(0.051)(0.051) \approx 0.00087$ Human: $\frac{1}{3}(0.400)(0.700) \approx 0.0933$ AI: $\frac{1}{3}(0.051)(0.550) \approx 0.00935$

Normalizing (dividing by the sum ≈ 0.1035):

$$P(\text{Legit}) \approx 0.8\%, \qquad P(\text{Human}) \approx 90.2\%, \qquad P(\text{AI}) \approx 9.0\%$$

4.4 Final Two-Step Decision Rule

Step 1 (Legitimate vs. Phishing):

$$P(\text{Phishing}) = P(\text{Human}) + P(\text{AI}) = 90.2\% + 9.0\% = 99.2\% \ (\geq 0.50 \Rightarrow \text{Phishing})$$

Step 2 (Human vs. AI, only if Phishing):

$$P(\text{AI} \mid \text{Phishing}) = \frac{9.0}{9.0 + 90.2} \approx 9.1\% \ (< 0.50 \Rightarrow \text{Human-Written})$$

5 End-to-End Summary

One email, carried through every stage of the current pipeline:

1. **Training time (once):** Kruskal–Wallis confirmed burstiness differs across classes ($H \approx 9.55$); Chi-Square confirmed urgency-keyword use differs across classes ($\chi^2 \approx 78.8$). Both passed FDR correction and their effect-size checks, so both were locked into the selected feature set along with their trained bin edges / frequency tables.
2. **New email arrives:** its burstiness value lands in the *High* bin; it contains an urgency keyword.
3. **Classification:** Model 1 (Rank-Binned NB) and Model 2 (Categorical NB) independently score the email against the stored training statistics; Meta-Naive Bayes combines both scores.
4. **Decision:** $P(\text{Phishing}) = 99.2\% \Rightarrow$ Phishing; $P(\text{AI} \mid \text{Phishing}) \approx 9.1\% \Rightarrow$ **Human-Written**.

Every number in this final verdict traces back to a specific, auditable test result from Phase 1 and a specific, auditable probability lookup from Phase 2 — which is exactly the "full auditability" property the pipeline is designed around.